



PROJECT NUMBER

CD

SHEET NUMBER

A. SEE DRAWING E-01 FOR ABBREVIATIONS, SYMBOLS, GENERAL NOTES AND DEVICE MOUNTING HEIGHT OF WALL MOUNTED DEVICES. UCON

B. SOLE PRODUCT OR MANUFACTURER/SOURCE: WHERE A SINGLE PRODUCT OR MANUFACTURER/SOURCE IS NAMED, SUBJECT TO COMPLIANCE WITH REQUIREMENTS, COMPARABLE PRODUCTS WILL BE CONSIDERED.

C. SEE T AND Y SERIES PLANS FOR ADDITIONAL INFORMATION AND REQUIREMENTS.

D. COORDINATE ALL DEVICE LOCATIONS AND MOUNTING HEIGHTS WITH ARCHITECTURAL DRAWINGS PRIOR TO ROUGH-IN.

E. THE CONTRACTOR SHALL COORDINATE EXACT LOCATION OF EQUIPMENT POWER CONNECTIONS WITH EQUIPMENT INSTALLER PRIOR TO ROUGH-IN.

F. CONTRACTOR SHALL BE RESPONSIBLE FOR SUPPLYING ANY REQUIRED TRANSFORMER/POWER SUPPLIES FOR DOOR HARDWARE. COORDINATE WITH ARCHITECTURAL AND SECURITY DRAWINGS FOR DOOR HARDWARE. PROVIDE 120V POWER AS REQUIRED.

G. PROVIDE WIRING METHODS COMPLIANT WITH UL 1196 STANDARD FOR FIRE RESISTIVE CABLING.

H. VOLTAGE DROP SHALL BE LIMITED TO LESS THAN 3% FOR ALL BRANCH CIRCUITS AND 2% FOR ALL FEEDERS.

I. X-RAY EXISTING FLOORS TO COORDINATE ALL WIRING DEVICE AND CORE DRILL LOCATIONS WITH ARCHITECTURAL DRAWINGS PRIOR TO ROUGH-IN LOCATION TO BE DETERMINED IN CONSULTATION WITH FURNITURE SELECTION.

J. CONTRACTOR SHALL COORDINATE AND FIELD VERIFY REQUIREMENT OF NEUTRAL CONDUCTOR FOR ALL EQUIPMENT AND PROVIDE NEUTRAL CONDUCTOR AS NECESSARY.

K. FOR WORKSTATION CIRCUITS SHARING A NEUTRAL, PROVIDE MULTI-POLE CIRCUIT BREAKER. IE DEVICES ARE NOT ACCEPTABLE.

L. TRIPPING OF GFCI RECEPTACLES SHALL NOT AFFECT THE DOWNSTREAM RECEPTACLES.

M. NO MORE THAN 3 CIRCUITS MAY SHARE A HOMERUN. CIRCUITS SHALL BE CONFIGURED WITH DEDICATED NEUTRAL CONDUCTORS.

N. CIRCUITS FEEDING AV EQUIPMENT SHALL SHARE SAME PHASE.

O. ALL UNUSED WALL OUTLETS TO BE COVERED WITH BLANK PLATE. ALL BLANK PLATE COLORS TO MATCH WALL COLOR.

P. SEE ANY COMMUNICATIONS PATHWAY PLAN AND AV DRAWINGS. FOR ADDITIONAL CONDUIT TO BE PROVIDED BY ELECTRICAL CONTRACTOR.

Q. ALL CONDUIT ROUTED WITHIN SLAB SHALL COMPLY WITH GUIDELINES LISTED IN STRUCTURAL DRAWINGS.

R. ELECTRICAL ROOMS IN ACCORDANCE WITH NEC 110.26.

S. OPEN CEILING AREAS ARE CONSIDERED PART OF THE ARCHITECTURAL DESIGN ELEMENT. COORDINATE ROUTING AND MOUNTING OF CONDUIT, CABLEING OR DEVICES WITH ARCHITECT PRIOR TO INSTALLATION.



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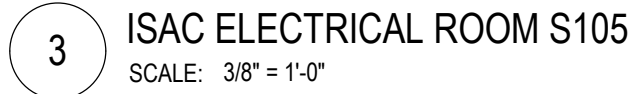
ISSUED FOR	REV	DATE
BID SET		04MAY2021
CONSTRUCTION DOCUMENTS		15APR2021
DESIGN DEVELOPMENT		23DEC2020
SCHEMATIC DESIGN		19OCT2020

Exp 10-31-21

ELECTRICAL - ENLARGED ROOMS

CD
SHEET NUMBER

E2.3



Short-Circuit Calculations

The following calculations are based on the "Point-by Point" method where:

$$I_{sc} = I_{sc} \times M$$

$$M = 1/(1+f)$$

$$f = 1.732 \times L \times I$$

$$C \times E$$

$$XFMR: IP(sca) = IP(sca) \times V_p \times \frac{Z}{100,000 \times KVA}$$

$$IS(sca) = \frac{V_p \times M \times IS(sca)}{V_s}$$

SOURCE FAULT VALUE DERIVED FROM PREVIOUSLY ENGINEERED PLANS

Fault Point	Fault Point Description	Source (Fault Point)	Source I (amps)	Conduit Type	Wire/Bus Size	Wire/Bus Type	'C' value	E (volts)	L (length)	X'FMR KVA	X'FMR Z	f	M	Isc
1	EXISTING MSB-SOUTH	25044												25044
2	ISACHA	25044	M	1	Set(s) of 500 KCML	AL	18755	480	197			0.849	0.51	11647
3	T-ISAC-LAWA PRI	12848	M	1	Set(s) of 2/0	AL	7186	480	12			0.077	0.93	11925
4	T-ISAC-LAWA SEC	11925	M		Set(s) of		#N/A	208		75	5.2	6.866	0.13	3498
5	T-ISAC-SA PRI	12848	M	1	Set(s) of 2/0	AL	7186	480	8			0.052	0.96	12218
6	T-ISAC-SA SEC	12218	M		Set(s) of		#N/A	208		75	5.2	7.034	0.12	3509

- 1) ELECTRICAL EQUIPMENT AIC RATING SHALL EXCEED CALCULATED FAULT VALUE
2) ELECTRICAL EQUIPMENT FED FROM A TRANSFORMER SECONDARY RESULTING IN LESS THAN 10000 IEC SHALL BE RATED 10K AIC

Ex Switchboard: MSB-SOUTH

Location: ELEC E-1-S11

Supply From: MOUNTING: ENCLOSURE:

Volts: 480Y/277

Phases: 3

Wires: 4

A.I.C. Rating: 65,000

Mains Type: MCB

Mains Rating: 4000 A

MCB Rating: 4000 A

CKT	Circuit Description	# of Poles	Frame Size	Trip Rating	Load	Remarks
1	EXISTING LOAD PER PREVIOUSLY ENGINEERED PLANS	3	0 A	0 A	1708536 VA	
2	ISACHA	3	400 A	300 A	151320 VA	
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						
Total Conn. Load:					1859856 VA	
Total Amps:					2237 A	

Load Classification	Connected Load	Demand Factor	Estimated Demand	Panel Totals	
Lighting	2997 VA	125.00%	3746 VA		
Power	1788332 VA	100.00%	1788332 VA	Total Conn. Load:	1859856 VA
Receptacle	18000 VA	77.78%	14000 VA	Total Est. Demand:	1862536 VA
Mechanical Equipment	50527 VA	111.74%	56458 VA	Total Conn.:	2237 A
				Total Est. Demand:	2240 A

Notes:

SWITCHBOARD SCHEDULE IS NOT INDICATIVE OF EXISTING ELECTRICAL EQUIPMENT DEVICE CONFIGURATION AND IS SHOWN FOR LOAD CALCULATION PURPOSES ONLY

Short-Circuit Calculations

The following calculations are based on the "Point-by Point" method where:

$$I_{sc} = I_{sc} \times M$$

$$M = 1/(1+f)$$

$$f = 1.732 \times L \times I$$

$$C \times E$$

$$XFMR: IP(sca) = IP(sca) \times V_p \times \frac{Z}{100,000 \times KVA}$$

$$IS(sca) = \frac{V_p \times M \times IS(sca)}{V_s}$$

SOURCE FAULT VALUE DERIVED FROM PREVIOUSLY ENGINEERED PLANS

Fault Point	Fault Point Description	Source (Fault Point)	Source I (amps)	Conduit Type	Wire/Bus Size	Wire/Bus Type	'C' value	E (volts)	L (length)	X'FMR KVA	X'FMR Z	f	M	Isc
1	EXISTING MSB-NORTH	25044												22000
2	UCCSHA	22000	M	1	Set(s) of 500 KCML	AL	18755	480	210			0.889	0.53	11647
3	T-UCCS-LA/LC PRI	11647	M	1	Set(s) of 300 KCML	AL	13909	480	12			0.036	0.97	11240
4	T-UCCS-LA/LC SEC	11240	M		Set(s) of		#N/A	208		150	3	1.867	0.35	9048
5	T-UCCS-SA PRI	11647	M	1	Set(s) of 6	CU	2425	480	8			0.139	0.88	10229
6	T-UCCS-SA SEC	10229	M		Set(s) of		#N/A	208		30	4.1	11.609	0.08	1872

- 1) ELECTRICAL EQUIPMENT AIC RATING SHALL EXCEED CALCULATED FAULT VALUE
2) ELECTRICAL EQUIPMENT FED FROM A TRANSFORMER SECONDARY RESULTING IN LESS THAN 10000 IEC SHALL BE RATED 10K AIC

Ex Switchboard: MSB-NORTH

Location: ELEC E-1-S21

Supply From: MOUNTING: ENCLOSURE:

Volts: 480Y/277

Phases: 3

Wires: 4

A.I.C. Rating: 22,000

Mains Type: MAIN FUSE

Mains Rating: 2500 A

MCB Rating: 2000 A

CKT	Circuit Description	# of Poles	Frame Size	Trip Rating	Load	Remarks
1	EXISTING LOAD PER PERVIOUSLY ENGINEERED PLANS	3	0 A	0 A	1344558 VA	
2	UCCSHA	3	400 A	300 A	147667 VA	
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						
Total Conn. Load:					1492225 VA	
Total Amps:					1795 A	
Load Classification		Connected Load	Demand Factor	Estimated Demand	Panel Totals	
Lighting		14625 VA	125.00%	18282 VA	Total Conn. Load: 1492225 VA Total Est. Demand: 1485702 VA Total Conn.: 1795 A Total Est. Demand: 1787 A	
Power		1429618 VA	100.00%	1429618 VA		
Receptacle		32440 VA	65.41%	21220 VA		
Mechanical Equipment		15542 VA	106.69%	16582 VA		
Notes:						
SWITCHBOARD SCHEDULE IS NOT INDICITIVE OF EXISTING ELECTRICAL EQUIPMENT DEVICE CONFIGURATION AND IS SHOWN FOR LOAD CALCULATION PURPOSES ONLY						

GENERAL SHEET NOTES

- A. SEE DRAWING E0.01 FOR ABBREVIATIONS, SYMBOLS, GENERAL NOTES.
- B. SOLE PRODUCT OR MANUFACTURER/SOURCE: WHERE A SINGLE PRODUCT OR MANUFACTURER/SOURCE IS NAMED, SUBJECT TO COMPLIANCE WITH REQUIREMENTS, COMPARABLE PRODUCTS WILL BE CONSIDERED.
- C. VOLTAGE DROP SHALL BE LIMITED TO LESS THAN 3% FOR ALL BRANCH CIRCUITS AND 2% FOR ALL FEEDERS.
- D. ALL PANELBOARDS SHALL BE EQUIPPED WITH BOLT ON BREAKERS ONLY.
- E. CONTRACTOR TO VERIFY THAT LUGS ON ALL ELECTRICAL EQUIPMENT CAN ACCEPT THE WIRE SIZES INDICATED ON SCHEDULE AND PLANS.



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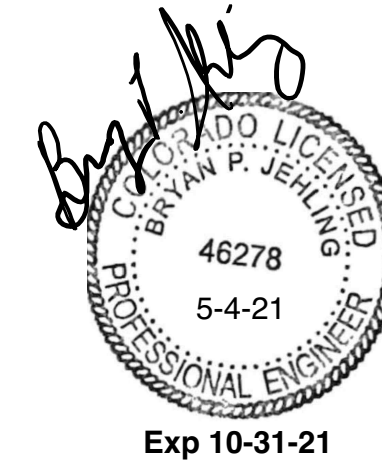
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SHEET KEYNOTES

- E501 PROVIDE NEW CIRCUIT BREAKER IN EXISTING ELECTRICAL EQUIPMENT. NEW CIRCUIT BREAKER SHALL MATCH EXISTING CIRCUIT BREAKER TYPE AND MANUFACTURER. SHORT CIRCUIT WITHSTAND RATING SHALL MEET OR EXCEED EXISTING DEVICE RATING.
- E502 PROVIDE 1.8KVA EMERGENCY LIGHTING INVERTER, MYERS ILLUMINATOR EM #6-EM42-S-B-D-20-03-TW, 69KVAIC, OR EQUAL. INSTALL PER EQUIPMENT MANUFACTURER SPECIFICATIONS. INCLUDE STARTUP, TRAINING AND EXTENDED WARRANTY.
- E503 PROVIDE 3.75KVA EMERGENCY LIGHTING INVERTER, MYERS ILLUMINATOR SUPERNOVA #6-I-03-S-B-A-20-03-T, 69KVAIC, OR EQUAL. INSTALL PER EQUIPMENT MANUFACTURER SPECIFICATIONS. INCLUDE STARTUP, TRAINING AND EXTENDED WARRANTY.

BID SET 04MAY2021
CONSTRUCTION DOCUMENTS 15APR2021
DESIGN DEVELOPMENT 23DEC2020

SEALS AND SIGNATURES



FEEDER SCHEDULE

INDICATES ALUMINUM CONDUCTORS

- A110-3 3#2/0 + 1#4 G, 1 1/2" C
- A125T 4#2/0 + 1#10 G, 2" C
- A200T 4#250KCMIL + 1#2 G, 2 1/2" C
- A225 4#300KCMIL + 1#2 G, 2 1/2" C
- A225-3 3#300KCMIL + 1#2 G, 2 1/2" C
- A225TB 4#300KCMIL + 1#30 G, 2 1/2" C
- A300 4#500KCMIL + 1#2 G, 3" C
- T30 30KVA TRANSFORMER, 480-208Y/120V, 3PH
PRIMARY FEEDER, 3#6 + 1#10 G, 3/4" C
SECONDARY FEEDER, 4#1 + 1#6 G, 1 1/2" C
BONDING JUMPERS & GROUNDING ELECTRODE CONDUCTOR, #6

SHEET TITLE

ELECTRICAL DIAGRAMS & CALCULATIONS

PROJECT NUMBER

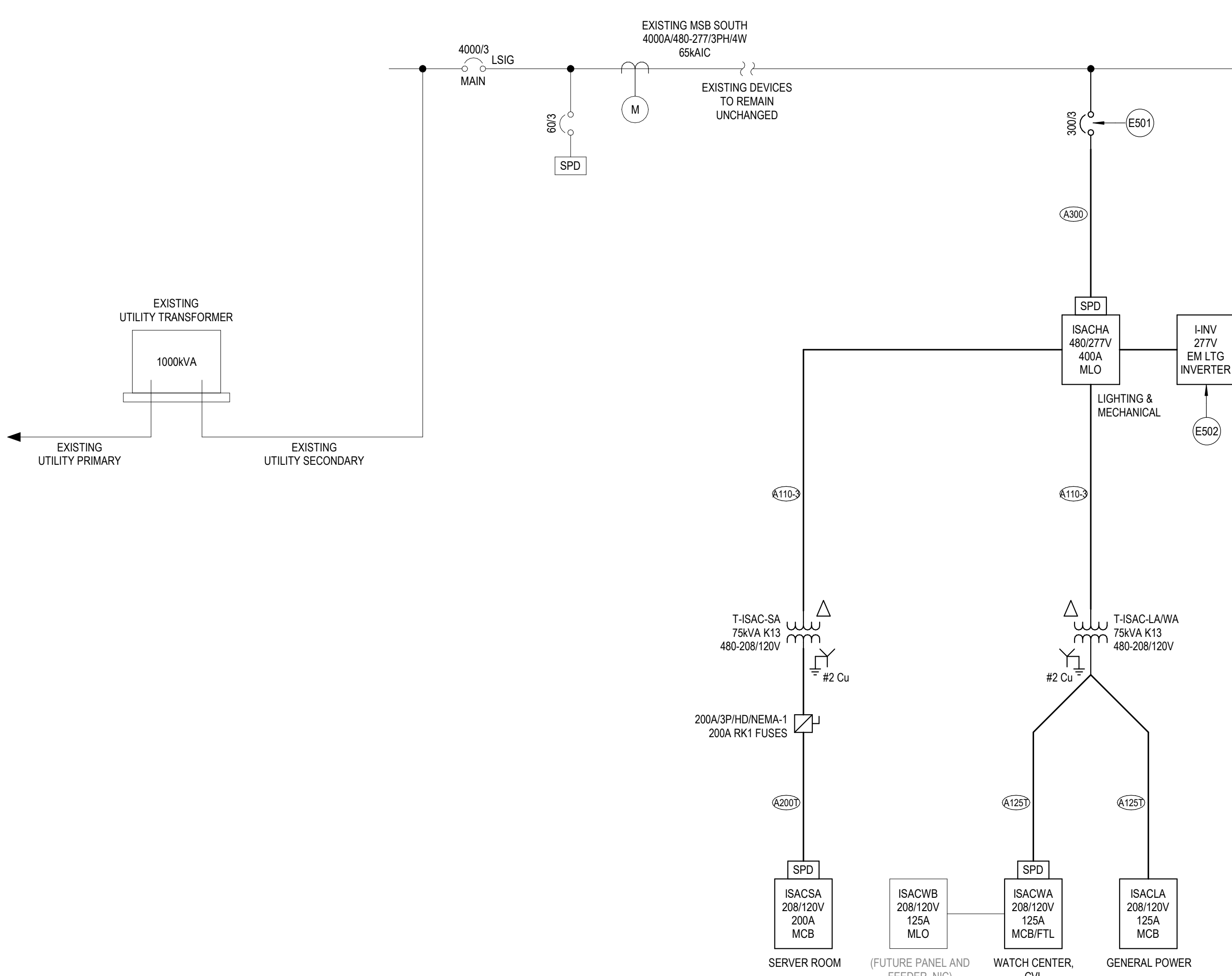


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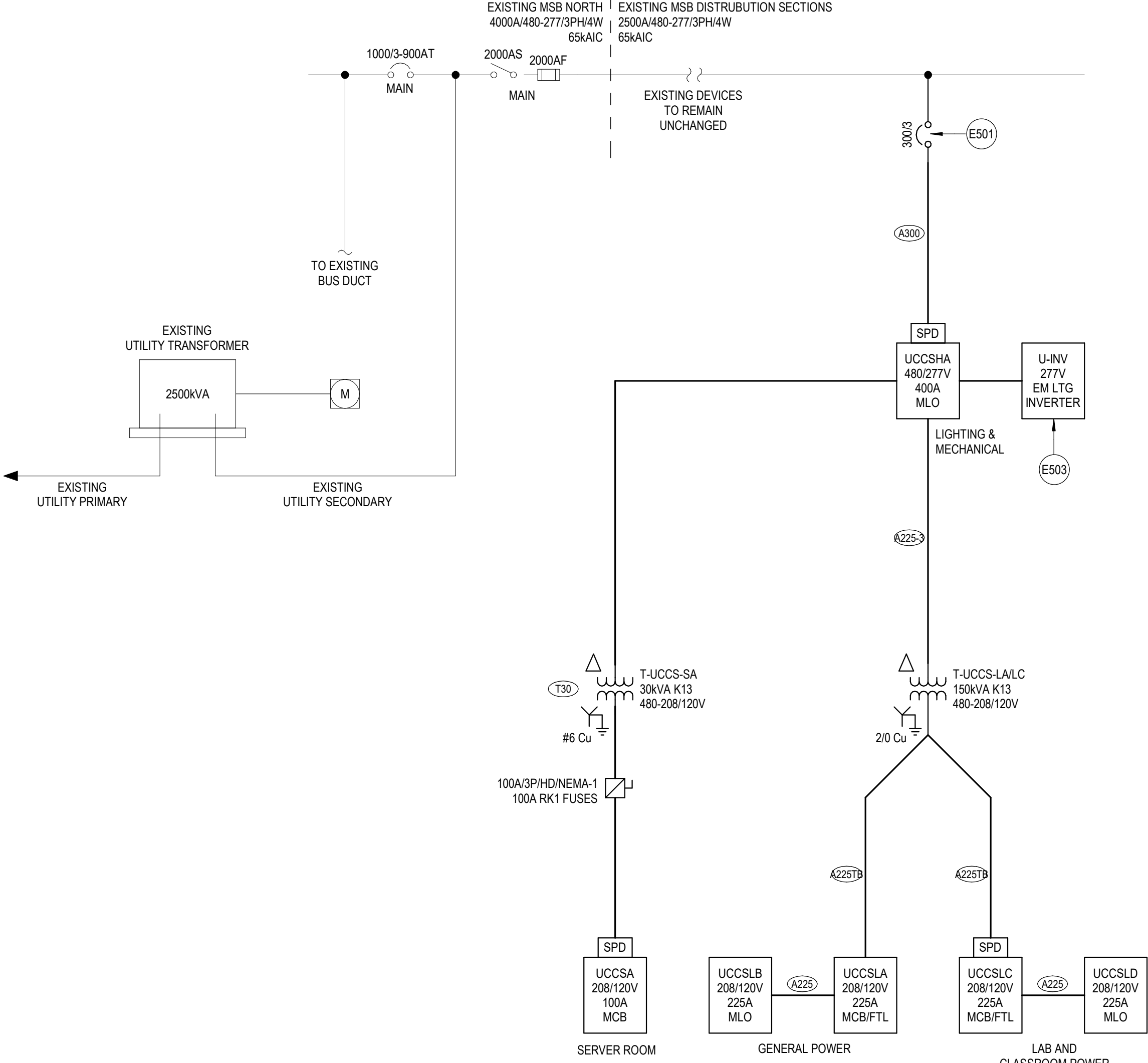
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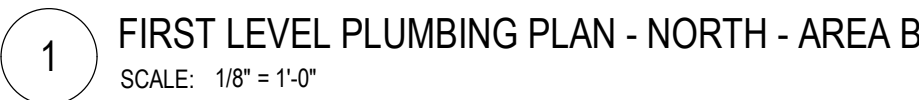
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SOUTH BUILDING - ISAC

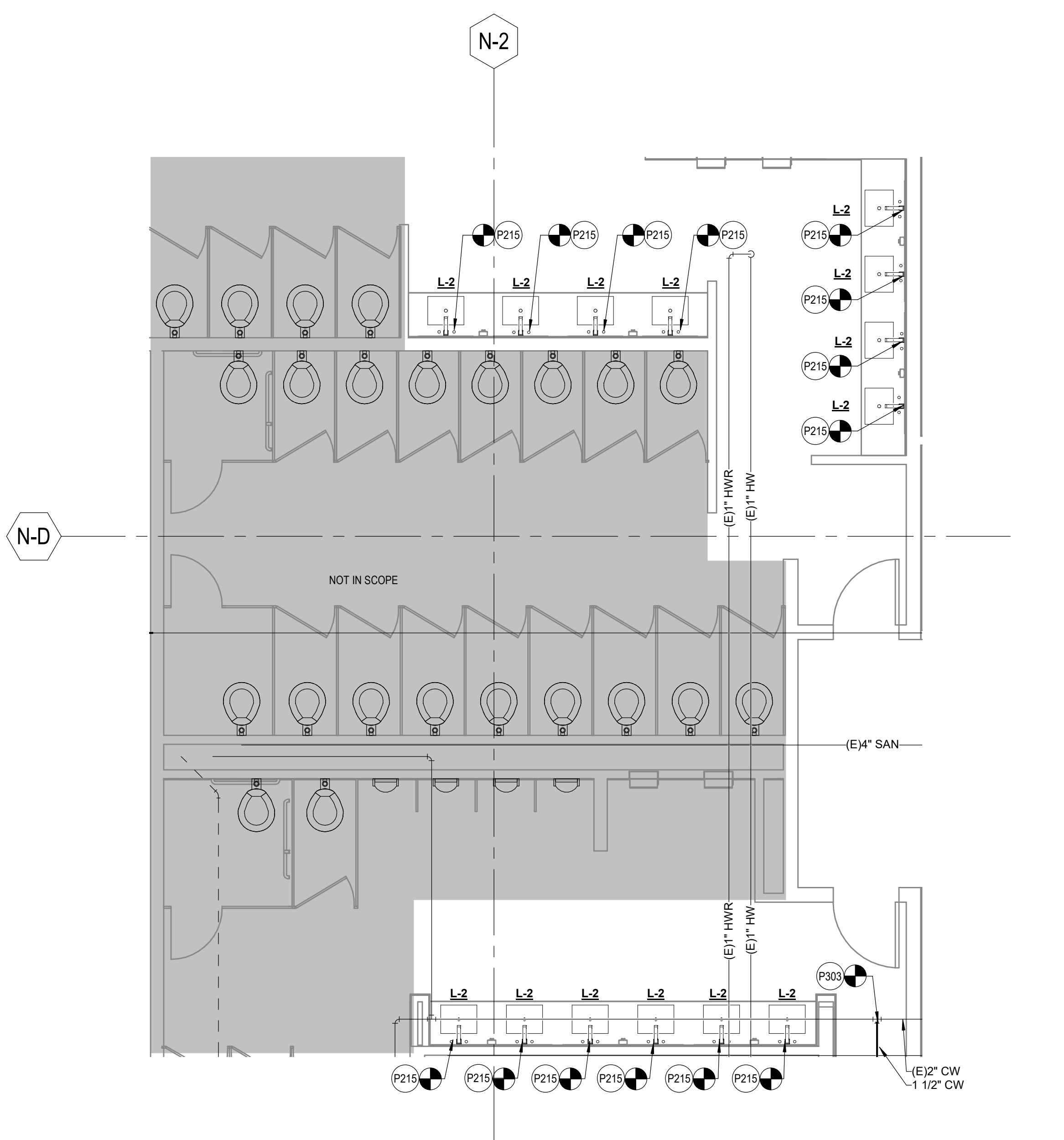


NORTH BUILDING - UCCS

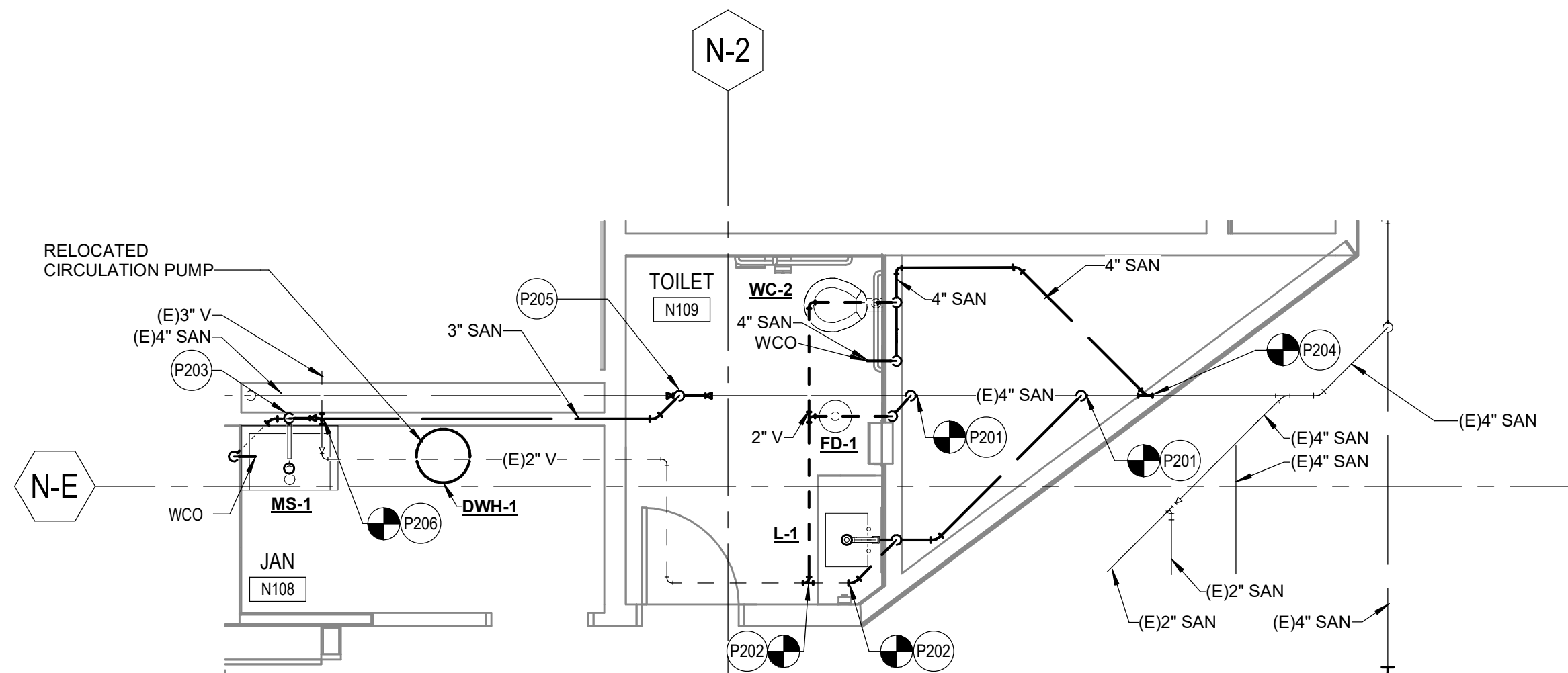




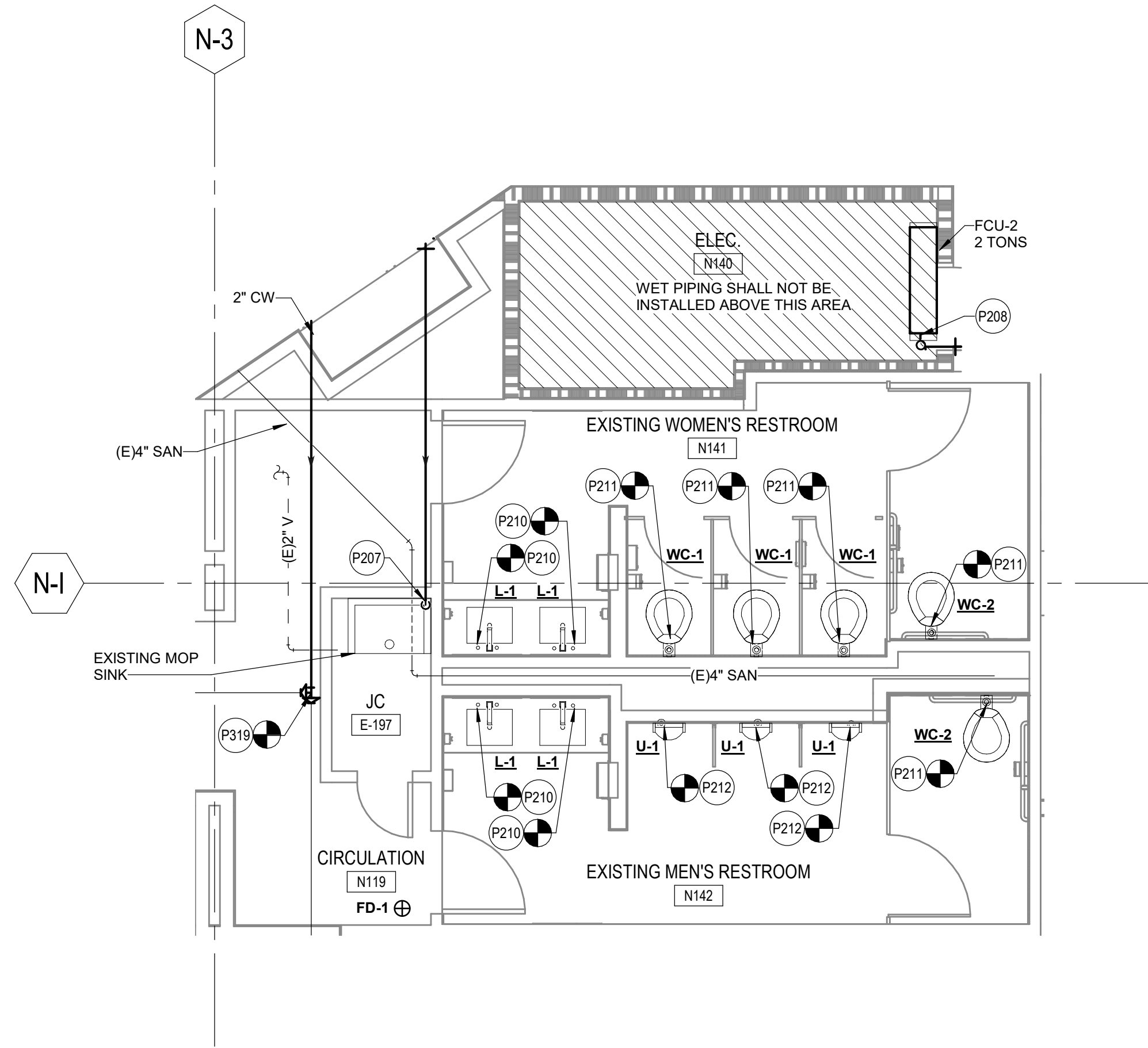
P2.1.B



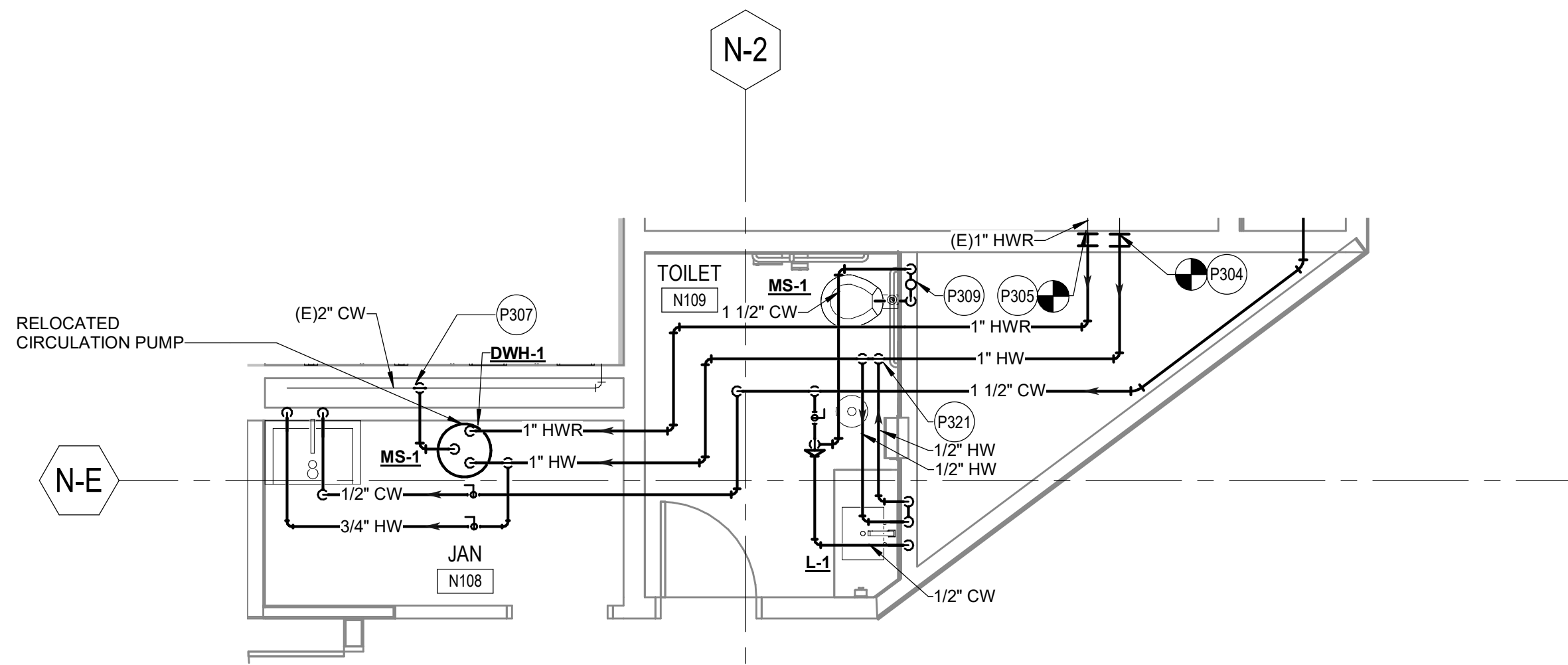
4 DEDUCT ALTERNATE - FIRST LEVEL ENLARGED PLUMBING PLAN - NORTH - AREA B
SCALE: 1/4" = 1'-0"



3 FIRST LEVEL ENLARGED PLUMBING SAN/VENT PLAN - NORTH - AREA B
SCALE: 1/4" = 1'-0"



2 FIRST LEVEL ENLARGED PLUMBING PLAN - NORTH - AREA A
SCALE: 1/4" = 1'-0"



1 FIRST LEVEL ENLARGED PLUMBING WATER PLAN - NORTH - AREA B
SCALE: 1/4" = 1'-0"

GENERAL SHEET NOTES

- ALL PIPING EXPOSED TO SUNLIGHT MUST BE METALLIC.
- SLOPE SANITARY WASTE PIPE 3" OR LARGER AT 1/8" PER FOOT (1 PERCENT) UNLESS OTHERWISE NOTED.
- SLOPE SANITARY WASTE PIPE 2-1/2" AND SMALLER AT 1/4" PER FOOT (2 PERCENT) UNLESS OTHERWISE NOTED.
- SLOPE CONDENSATE DRAIN AT 1/8" PER FOOT (1 PERCENT) UNLESS OTHERWISE NOTED.
- ALL HORIZONTAL DRAINS SHALL BE PROVIDED WITH CLEANOUTS LOCATED NO MORE THAN 100 FEET APART.
- CLEANOUTS SHALL BE INSTALLED AT THE BASE OF ALL STACKS AND EACH CHANGE OF DIRECTION GREATER THAN 45 DEGREES IN THE BUILDING SEWER, BUILDING DRAIN, AND HORIZONTAL WASTE OR SOIL LINES. WHERE MORE THAN ONE CHANGE IN DIRECTION OCCURS IN A RUN OF PIPING ONLY ONE CLEANOUT SHALL BE REQUIRED FOR EACH 50 FEET OF DEVELOPED LENGTH OF THE DRAINAGE PIPING FOR PIPING 4 INCHES AND SMALLER AND 100 FEET MINIMUM INTERVALS FOR PIPING LARGER THAN 4 INCHES.
- ALL FLOOR DRAINS AND FLOOR SINKS SHALL BE PROVIDED WITH TRAP GUARD.
- PROVIDE SHUT-OFF VALVE AT EACH EQUIPMENT AND FIXTURE CONNECTION.
- REFER TO SERVICE CONNECTION SCHEDULE FOR PIPE SIZE TO FIXTURES.
- SEE STRUCTURAL DRAWINGS FOR SLAB-ON-GRADE INFILL.



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SHEET KEYNOTES

- P201 CONNECT NEW 2" SANITARY FROM POINT OF CONNECTION TO EXISTING 4" SANITARY PIPE BELOW GRADE.
- P202 CONNECT NEW 2" VENT FROM POINT OF CONNECTION TO EXISTING 2" VENT.
- P203 2" VENT UP FROM MOP SINK.
- P204 CONNECT NEW 4" SANITARY FROM POINT OF CONNECTION TO 4" SANITARY PIPE BELOW GRADE.
- P205 CONNECT NEW 3" SANITARY FROM POINT OF CONNECTION TO 4" SANITARY PIPE BELOW GRADE.
- P206 CONNECT NEW 2" VENT FROM POINT OF CONNECTION TO 3" VENT.
- P207 TERMINATE 3/4" CONDENSATE DRAIN AT EXISTING MOP SINK WITH A MINIMUM 2" AIR GAP.
- P208 PROVIDE 3/4" TRAPPED AND VENTED CONDENSATE DRAIN. SEE DETAIL 309P1.
- P210 NEW LAVATORY FIXTURE AND FAUCET CONNECTION TO EXISTING ROUGH-INS.
- P211 NEW WATER CLOSET AND FLUSH VALVE CONNECTION TO EXISTING ROUGH-INS.
- P212 NEW URINAL AND FLUSH VALVE CONNECTION TO EXISTING ROUGH-INS.
- P215 ALTERNATE - REPLACING EXISTING LAVATORY SINK AND FAUCET WITH NEW L-2 SINK AND FAUCET.
- P303 CONNECT NEW 1 1/2" DOMESTIC COLD WATER FROM POINT OF CONNECTION TO EXISTING 2" PIPE ABOVE CEILING.
- P304 CONNECT NEW 1" DOMESTIC HOT WATER FROM POINT OF CONNECTION TO EXISTING 1" PIPE ABOVE CEILING.
- P305 CONNECT NEW 1" DOMESTIC HOT WATER RETURN FROM POINT OF CONNECTION TO EXISTING 1" PIPE ABOVE CEILING.
- P307 CONNECT NEW 1" DOMESTIC COLD WATER FROM POINT OF CONNECTION TO EXISTING 2" DOMESTIC COLD WATER.
- P309 PROVIDE WATER HAMMER ARRESTOR SIOUX CHIEF "SIZE A" WITH ACCESS PANEL.
- P319 CONNECT NEW 2" DOMESTIC COLD WATER FROM POINT OF CONNECTION TO EXISTING 3/4" DOMESTIC COLD WATER.
- P321 PROVIDE FLOW-SPLITTER UNIT, KEMPER MODEL 651-06 WITH SHUT-OFF VALVES.

SEALS AND SIGNATURES



SHEET TITLE
PLUMBING ENLARGED PLANS

PROJECT NUMBER

CD
SHEET NUMBER

12654.000

P4.1